2.5 Sensor Placement using field geometry

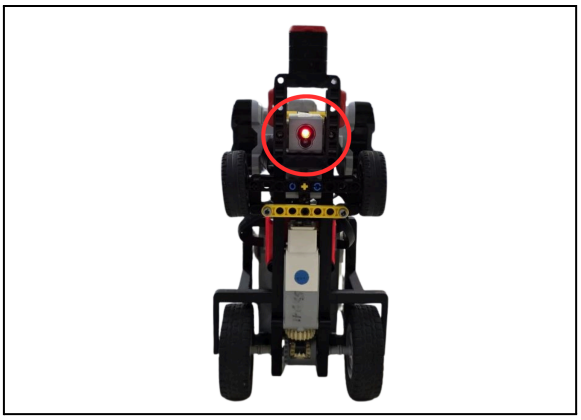
2.5.1 Ultrasonic Sensor Placement



Two Ultrasonic Sensors are mounted symmetrically at the front of the robot, facing opposite directions to monitor the left and right sides of the field. Both sensors are positioned approximately 8 cm above the ground, based on the chassis structure.

- **Wall Following:** Maintains a target distance of 270 mm from the selected wall using distance-based feedback control.
- **Field Geometry Adaptation:** Automatically selects the left or right wall based on the parking lot location.
- **Parking Lot Positioning:** Compares left and right distance readings to determine whether the robot is properly aligned with the parking area.
- **Mechanical Integration:** Dedicated chassis mounting points provide secure and organised sensor installation.

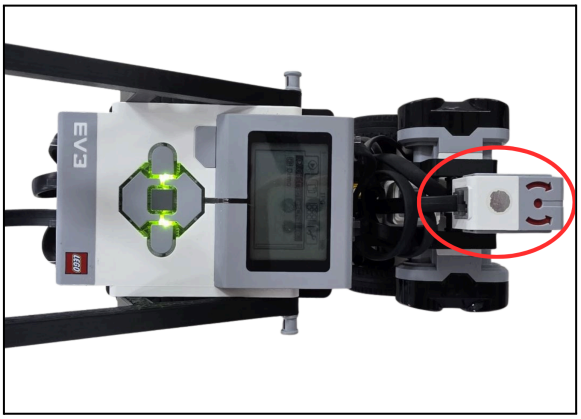
2.5.2 Color Sensor Placement



The EV3 Color Sensor is mounted at the front-centre of the robot, approximately 1 cm above the ground. This position allows early detection of coloured markings at the four corners of the square competition field, enabling timely 90° turning actions.

- **Field Geometry:** Two coloured lines are placed at each corner of the square field.
- **Direction Recognition:** Orange lines indicate clockwise movement, while blue lines indicate counterclockwise movement.
- **Colour Filtering:** The program detects only the colour corresponding to the selected direction and ignores the opposite colour.
- **Mechanical Integration:** Direct axle mounting connects the sensor to the steering assembly and maintains its front-centre position.

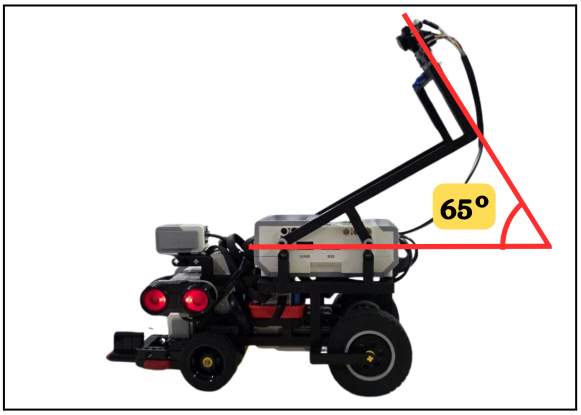
2.5.3 Gyro Sensor Placement



The Gyro Sensor is mounted directly above the Steering Motor, using a dedicated mounting position integrated into the chassis with LEGO axles or connectors.

- **Field Geometry:** Measures the robot's horizontal rotation (Yaw) to support 90° turns at the four corners of the square field.
- **Heading Control:** Uses cumulative heading measurements (90°, 180°, 270° and 360°) to maintain the robot's orientation.
- **Design Improvement:** Relocated from its previous position after unstable readings and sudden angle fluctuations were observed during testing.
- **Testing Outcome:** The new mounting position eliminated the recurring reading fluctuations during testing.

2.5.4 Pixy Camera Placement



The Pixy Camera is mounted at the rear-top of the robot, approximately 29 cm above the ground, with its lens facing forward.

- **Field Geometry:** The elevated position provides a forward view of the competition field, allowing the robot to observe pillars and the parking area.
- **Pillar Recognition:** Identifies red and green pillars and extracts their pixel coordinates for distance estimation using the Pythagorean theorem.
- **Mechanical Integration:** The custom 3D-printed bracket and LEGO connectors provide a dedicated camera mounting solution.